

Q6: Piecewise Functions

Continuity at Boundary

Check both pieces give same limit at the boundary point (x_0, y_0) :

$$\lim_{\text{both sides}} f = f(x_0, y_0)$$

Useful bound: $\left| \frac{x^2 - y^2}{x^2 + y^2} \right| \leq 1$ (squeeze theorem at origin).

Partial Derivatives at Boundary

Must compute **from definition** using one-sided limits:

$$f_x(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

Check both $h \rightarrow 0^+$ and $h \rightarrow 0^-$. If they differ $\Rightarrow$ **DNE**.

Differentiability at Origin

If partial derivatives $= (a, b)$, check:

$$\lim_{(dx, dy) \rightarrow 0} \frac{|f(dx, dy) - f(0, 0) - a dx - b dy|}{\|(dx, dy)\|} = 0$$

Key trick: If $|f(dx, dy)| \leq C\|(dx, dy)\|^k$ for $k > 1$, then differentiable.

Clairaut's Theorem

$D_x D_y f = D_y D_x f$ at p if $D_x D_y f$ is **continuous** at p .

Warning: Mixed partials can be equal at origin even when discontinuous — verify by definition.

Q4/Q7: Vector Fields

Divergence & Curl (Recap)

$$\begin{aligned}\nabla \cdot F &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ \nabla \times F &= \begin{pmatrix} \partial_y F_3 - \partial_z F_2 \\ \partial_z F_1 - \partial_x F_3 \\ \partial_x F_2 - \partial_y F_1 \end{pmatrix}\end{aligned}$$

Source / Sink via Eigenvalues

- All eigenvalues $\text{Re}(\lambda) > 0 \Rightarrow$ **source**
- All eigenvalues $\text{Re}(\lambda) < 0 \Rightarrow$ **sink**
- Mixed $\Rightarrow$ **neither** (saddle-like)

For 2×2 : $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$.

Inverse Function Theorem

F invertible near p if $\det[DF(p)] \neq 0$. Then:

$$[DF^{-1}(F(p))] = [DF(p)]^{-1}$$

Q5: Critical Points

Hessian Classification (2×2)

$$D = \det H = f_{xx}f_{yy} - f_{xy}^2$$

D	f_{xx}	Type
> 0	< 0	Local max
> 0	> 0	Local min
< 0	—	Saddle
$= 0$	—	Inconclusive

Constrained Optimization (Boundary)

On segment $x = c$: substitute to get $g(y) = T(c, y)$, then solve $g'(y) = 0$.

Don't forget endpoints of the segment!

Q3(e): Steepest Ascent

On a surface $F = 0$, the tangent vector in $\mathbb{R}^4$ in the direction of steepest ascent of T is:

$$\mathbf{v} = (\nabla T, |\nabla T|^2) \in \mathbb{R}^4$$

Moving along this vector means $dT = |\nabla T|^2$.

Q8: Matrix-Valued Functions

Derivative of $G = F^2$

If $h(A) = A^2$, the total derivative is:

$$[Dh(A)](X) = AX + XA$$

By chain rule: $D_{\mathbf{v}}G = F(P) \cdot (D_{\mathbf{v}}F) + (D_{\mathbf{v}}F) \cdot F(P)$

Abstract/Linear Maps

If $L : V \rightarrow V$ is **linear**, then $[DL](v) = L(v)$.

Product rule: $D_{(a,b)}(xy^T) = xb^T + ay^T$.

Q2: Newton's Method & Polar

Partial Derivatives in Polar

$x = r \cos \theta$, $y = r \sin \theta$. By chain rule:

$$\begin{aligned}D_r F &= F_x \cos \theta + F_y \sin \theta \\ D_\theta F &= -F_x r \sin \theta + F_y r \cos \theta\end{aligned}$$

IFT Quick Reference

Goal	Formula
R as fn of (S, I, T)	$\frac{\partial R}{\partial I} = -\frac{F_I}{F_R}$
I as fn of (S, T, R)	$\frac{\partial R}{\partial I} = -\frac{F_I}{F_R}$
Slope of $y(x)$	$\frac{dy}{dx} = -\frac{F_x}{F_y}$

Tangent plane to $F = 0$ at p :

$$\nabla F(p) \cdot (\mathbf{x} - p) = 0.$$